

DRAEM - ICCV 2021

A Discriminatively Trained Reconstruction Embedding for Surface Anomaly Detection – Vitjan Zavrtanik, Matej Kristan, Danijel Skočaj

Christopher Roßbach¹

¹Friedrich-Alexander-Universität Erlangen-Nürnberg, Technische Fakultät

January 12, 2026

- Detection and localization of anomalies in the image space

Introduction

Surface Anomaly Detection

- Detection and localization of anomalies in the image space
- E.g. RGB drone images, medical scans, ...

- Detection and localization of anomalies in the image space
- E.g. RGB drone images, medical scans, ...
- OOD samples are often rare and diverse

- Detection and localization of anomalies in the image space
- E.g. RGB drone images, medical scans, ...
- OOD samples are often rare and diverse
- Collecting data for all possible anomalies is often infeasible

- Detection and localization of anomalies in the image space
- E.g. RGB drone images, medical scans, ...
- OOD samples are often rare and diverse
- Collecting data for all possible anomalies is often infeasible
- Data collection gets even harder when pixel-accurate localization is required

- Detection and localization of anomalies in the image space
- E.g. RGB drone images, medical scans, ...
- OOD samples are often rare and diverse
- Collecting data for all possible anomalies is often infeasible
- Data collection gets even harder when pixel-accurate localization is required
- Therefore, methods that only rely on normal data are preferred

Problem Statement

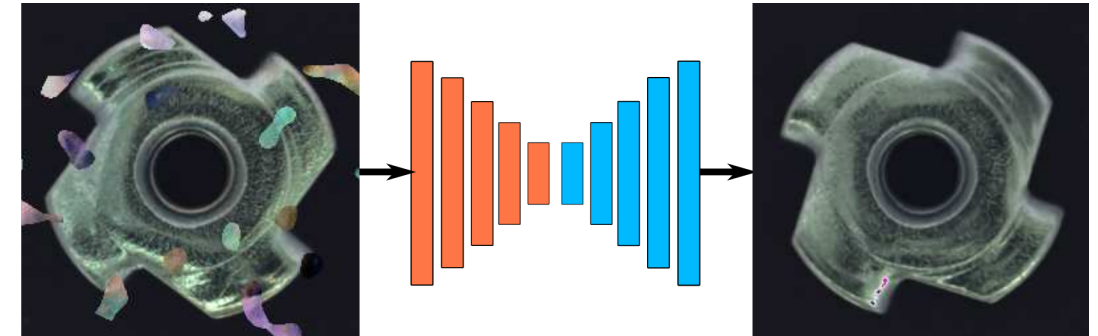
Current Situation and Related Work

- Two groups of methods in this area:

Problem Statement

Current Situation and Related Work

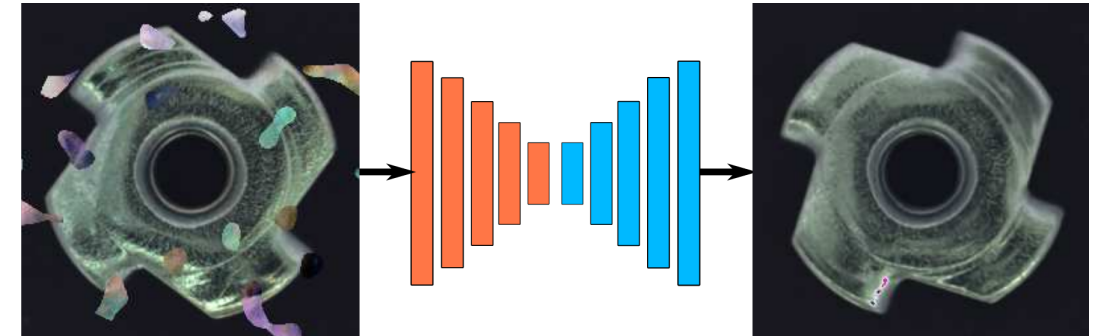
- Two groups of methods in this area:
1. **Reconstructive Methods** (Autoencoders, GANs) detect anomalies by measuring the reconstruction error



Problem Statement

Current Situation and Related Work

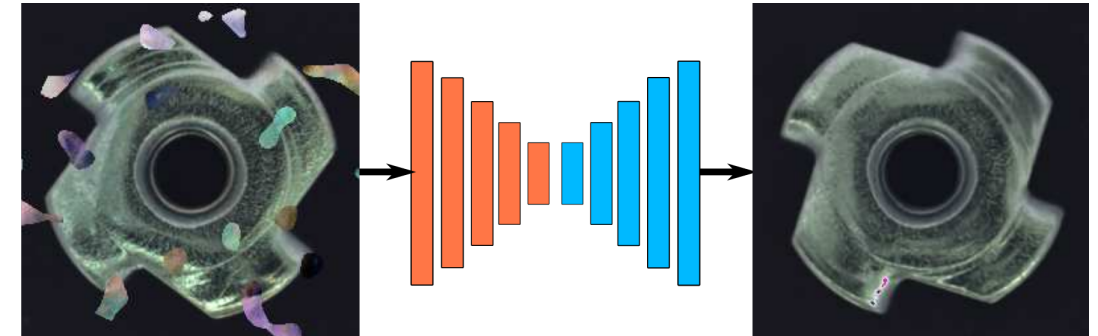
- Two groups of methods in this area:
 1. **Reconstructive Methods** (Autoencoders, GANs) detect anomalies by measuring the reconstruction error
 - Minor anomalies are often well reconstructed



Problem Statement

Current Situation and Related Work

- Two groups of methods in this area:
 1. **Reconstructive Methods** (Autoencoders, GANs) detect anomalies by measuring the reconstruction error
 - Minor anomalies are often well reconstructed
 2. **Discriminative Methods** detect distribution deviations on (deep) features (e.g. from CNNs)



Problem Statement

Current Situation and Related Work

- Two groups of methods in this area:
 1. **Reconstructive Methods** (Autoencoders, GANs) detect anomalies by measuring the reconstruction error
 - Minor anomalies are often well reconstructed
 2. **Discriminative Methods** detect distribution deviations on (deep) features (e.g. from CNNs)
 - Often suffer from blurry or coarse segmentation
 - Overfit to synthetic anomalies

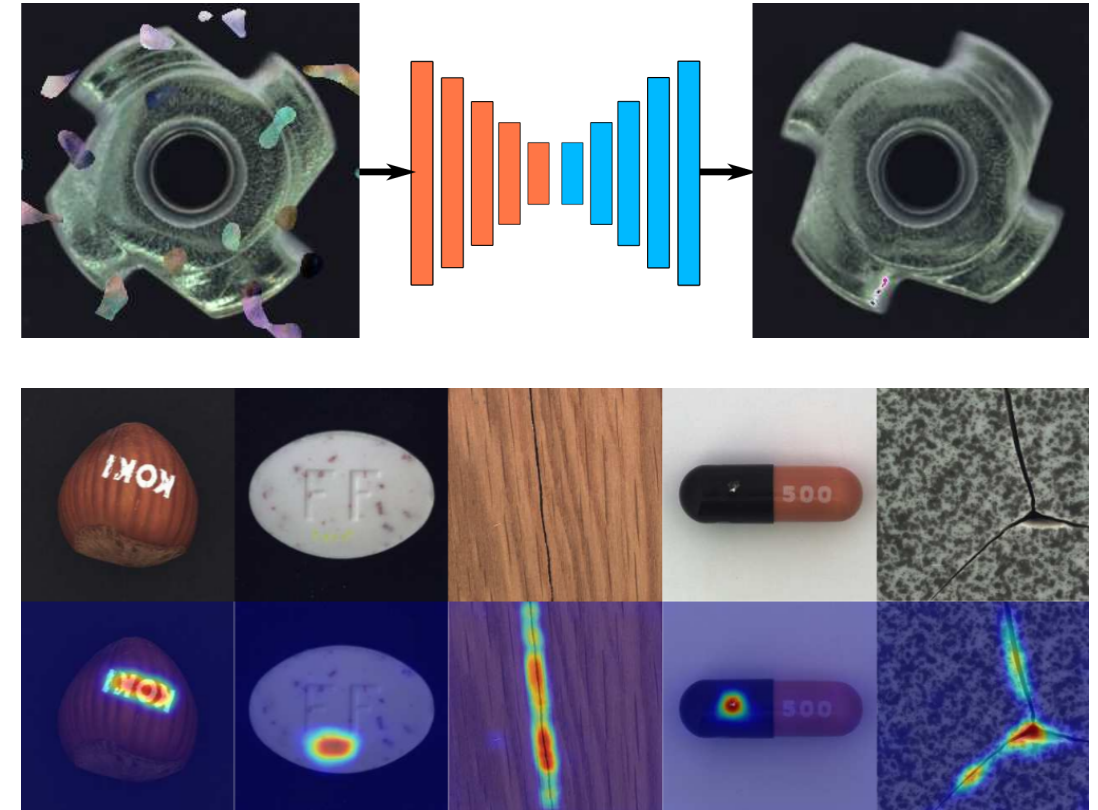
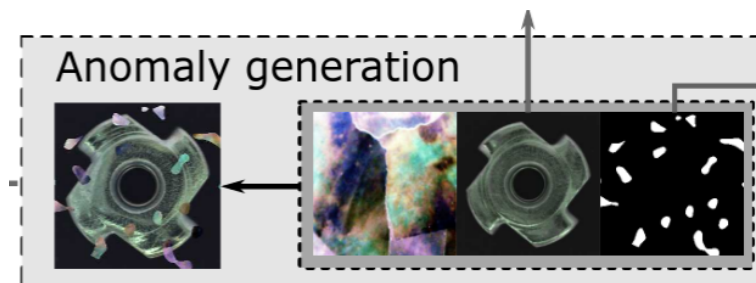


Figure: Blurry segmentation maps from discriminative methods.

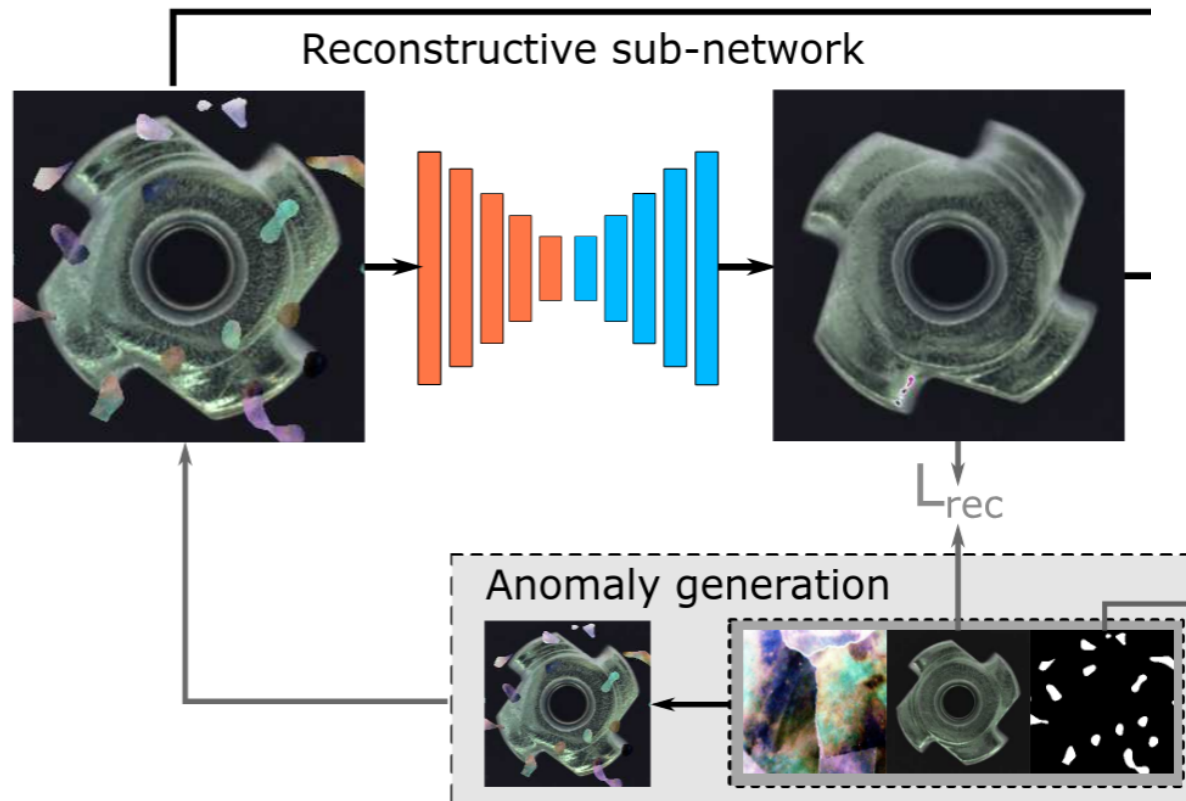
Framework Overview

How it Works



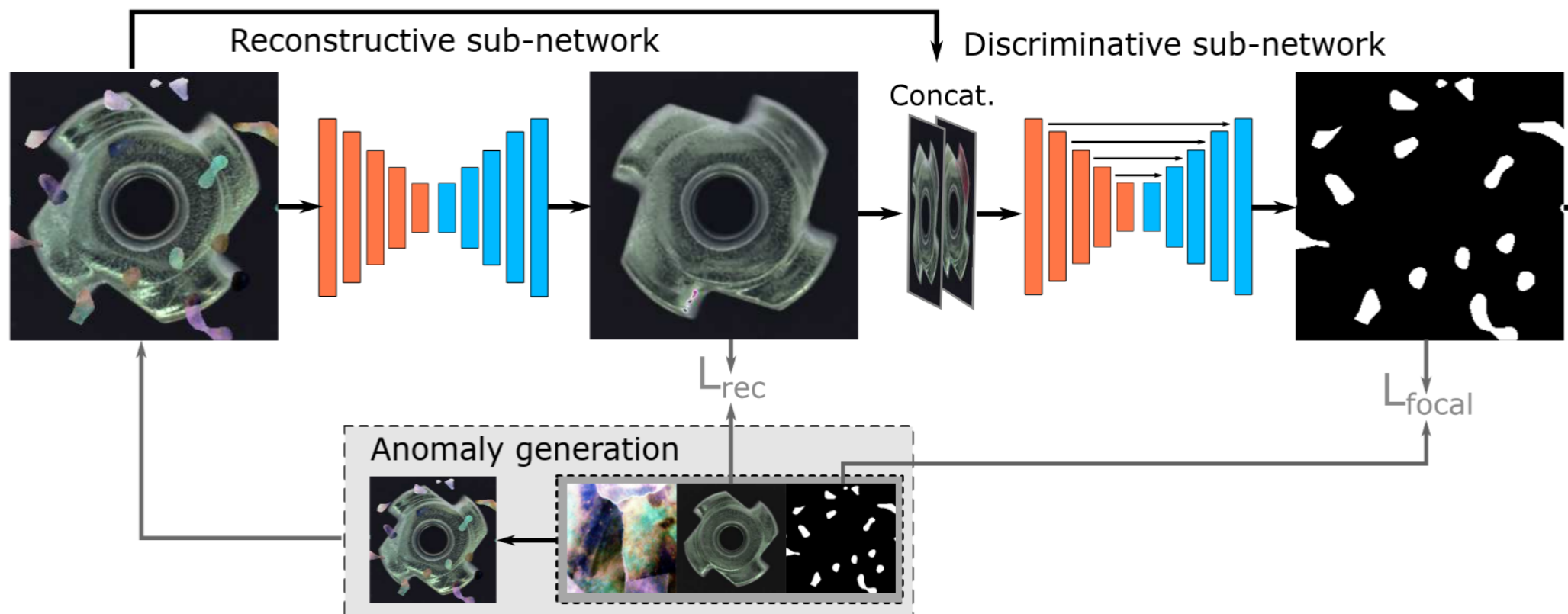
Framework Overview

How it Works



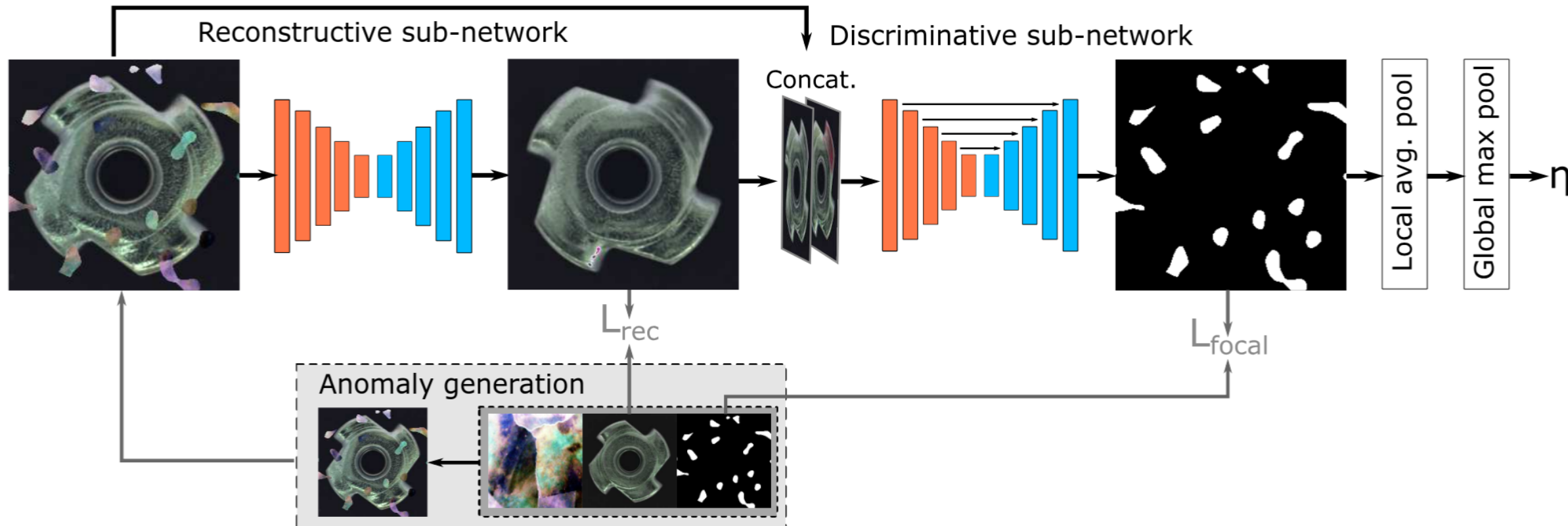
Framework Overview

How it Works



Framework Overview

How it Works



Performance Comparison

DRÆM vs State-of-the-Art

Method	Image AUROC ↑	Pixel AUROC ↑	Pixel AP ↑
US	87.7%	93.9%	45.5%
RIAD	91.7%	94.2%	48.2%
PaDim	<u>95.5%</u>	97.4%	<u>55.5%</u>
DRÆM	98.0%	<u>97.3%</u>	68.4%

Table: Results on the MVTec dataset.

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC ↑	TPR ↑	TNR ↑	CA ↑
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	99.0	96.5	99.4	98.5

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC ↑	TPR ↑	TNR ↑	CA ↑
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	99.0	96.5	99.4	98.5
Supervised				
CADN	—	—	—	89.1
Racki et al.	99.6	99.9	99.5	—
Lin et al.	99.0	99.4	99.9	—
Bozic et al.	100	100	100	100

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

Performance on DAGM Dataset

Unsupervised vs. Supervised Methods

Method	AUROC ↑	TPR ↑	TNR ↑	CA ↑
Unsupervised				
US	72.5	72.6	65.3	66.2
RIAD	78.6	79.2	69.1	70.4
PaDim	95.0	83.3	97.5	95.7
DRÆM	99.0	96.5	99.4	98.5
Supervised				
CADN	—	—	—	89.1
Racki et al.	99.6	99.9	99.5	—
Lin et al.	99.0	99.4	99.9	—
Bozic et al.	100	100	100	100

Table: DRÆM on DAGM dataset: outperforming unsupervised methods and on par with fully supervised ones.

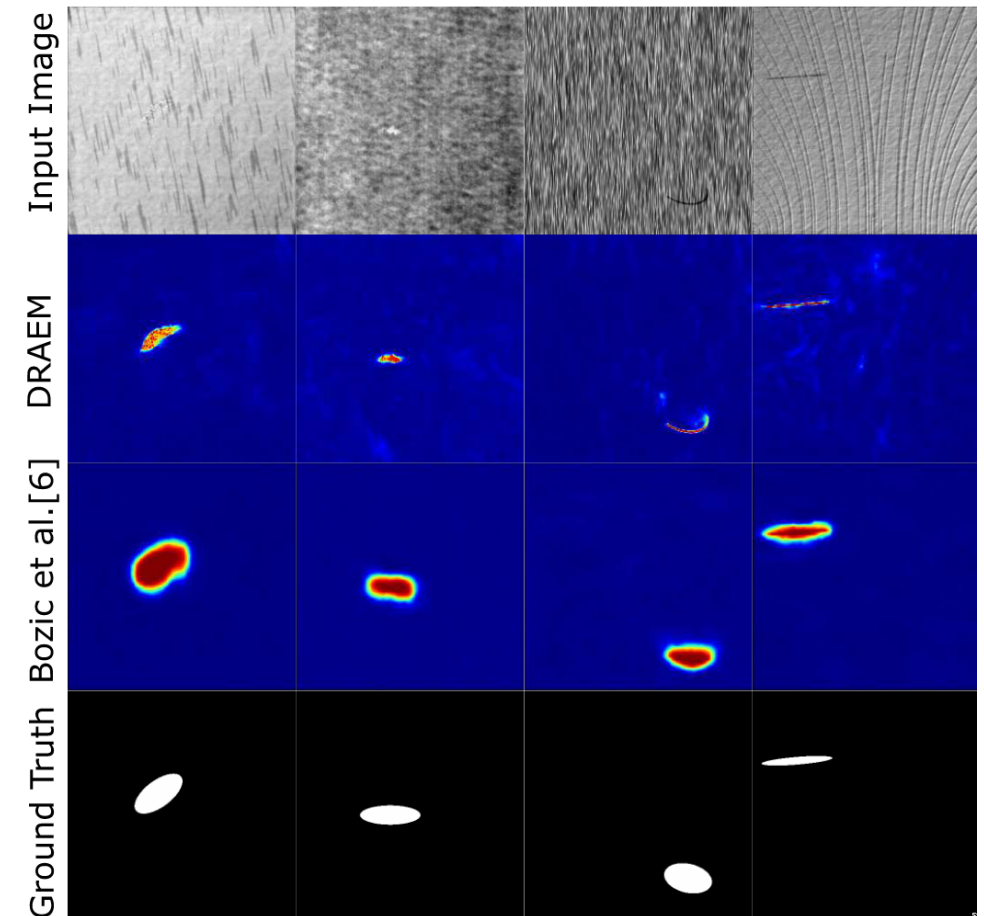


Figure: DRÆM vs supervised SOTA (Bozic et al.) on the DAGM dataset.

Key Findings & Application & Conclusion

The Value of DRÆM



- Outperforms unsupervised SOTA methods

Key Findings & Application & Conclusion

The Value of DRÆM

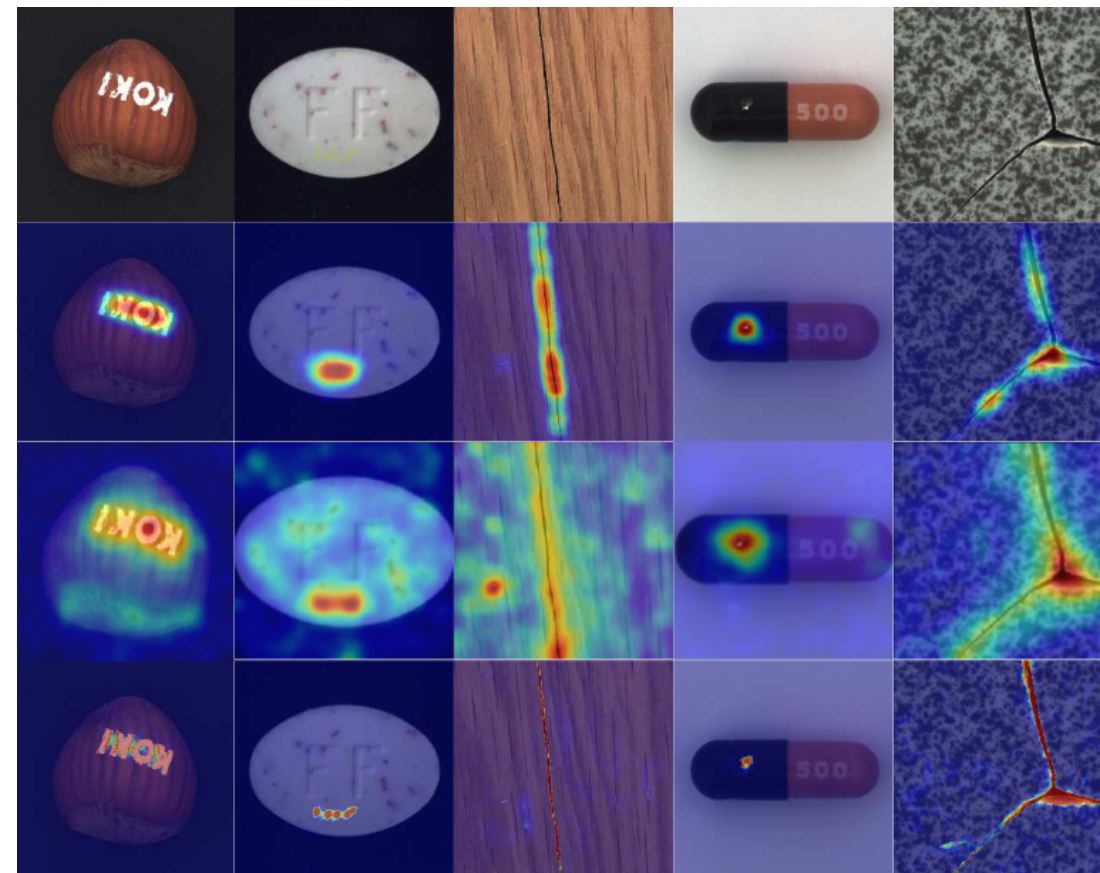


- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods

Key Findings & Application & Conclusion

The Value of DRÆM

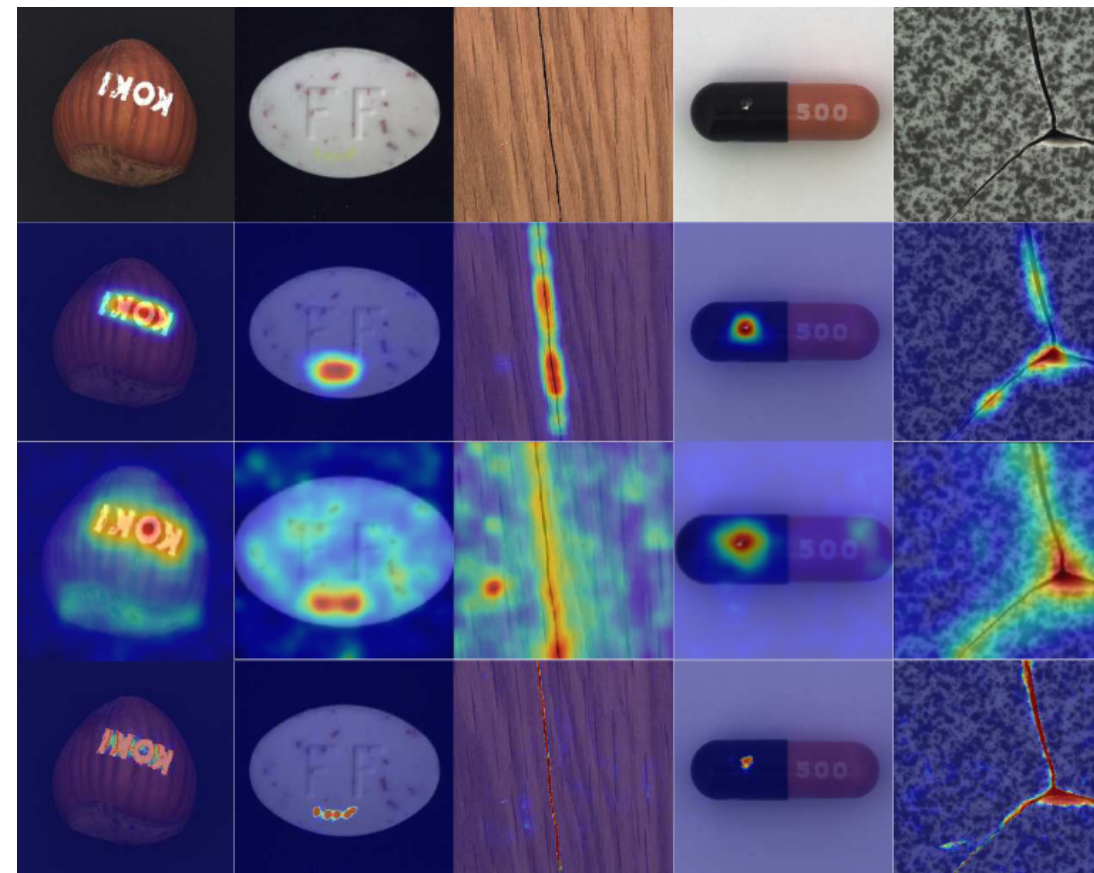
- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High-quality segmentation can help downstream tasks



Key Findings & Application & Conclusion

The Value of DRÆM

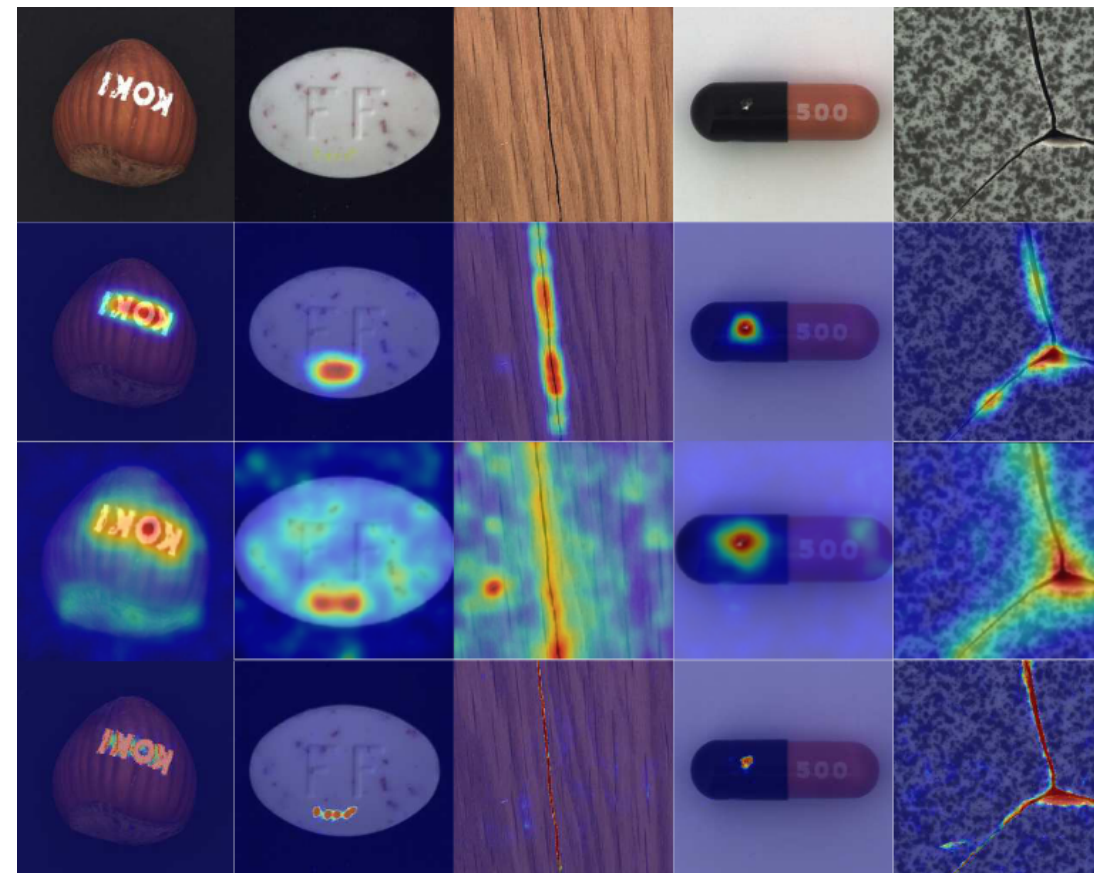
- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High-quality segmentation can help downstream tasks
 - Estimate the size of the impacted area



Key Findings & Application & Conclusion

The Value of DRÆM

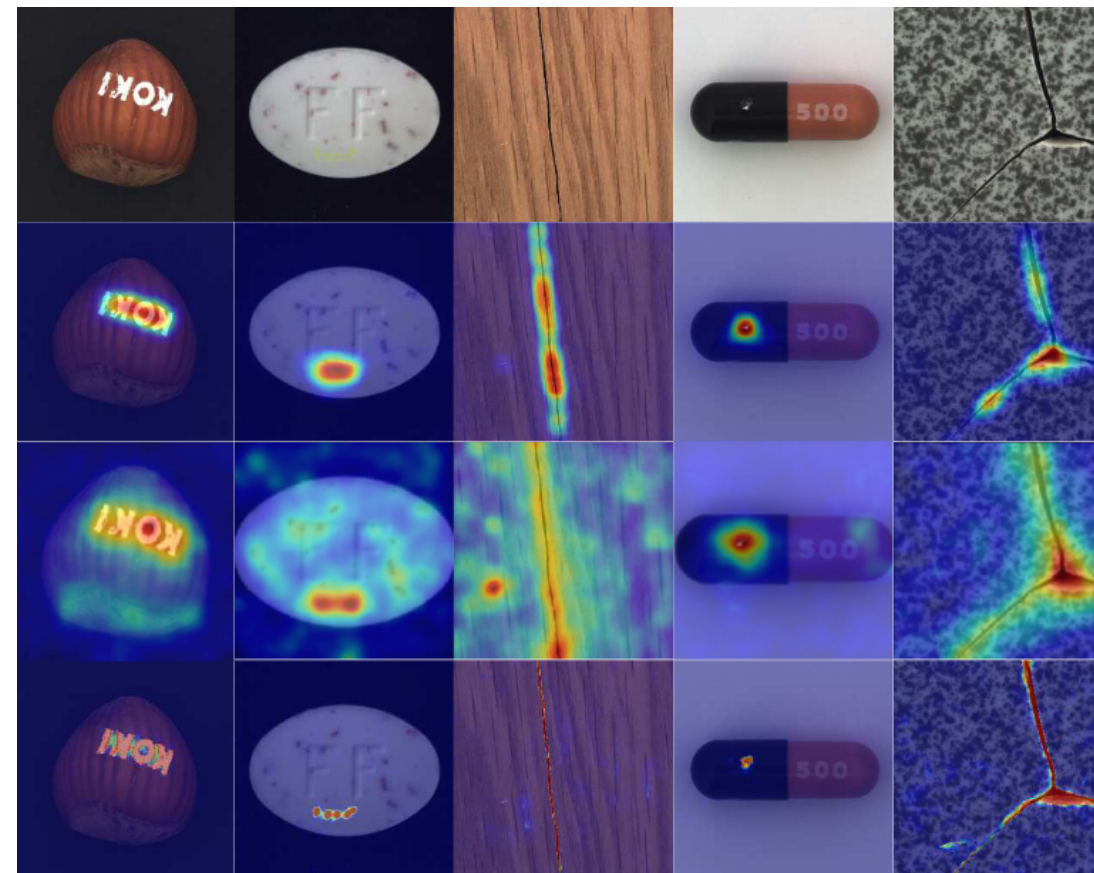
- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High-quality segmentation can help downstream tasks
 - Estimate the size of the impacted area
 - Distinguish between cracks, corrosion, or just paint



Key Findings & Application & Conclusion

The Value of DRÆM

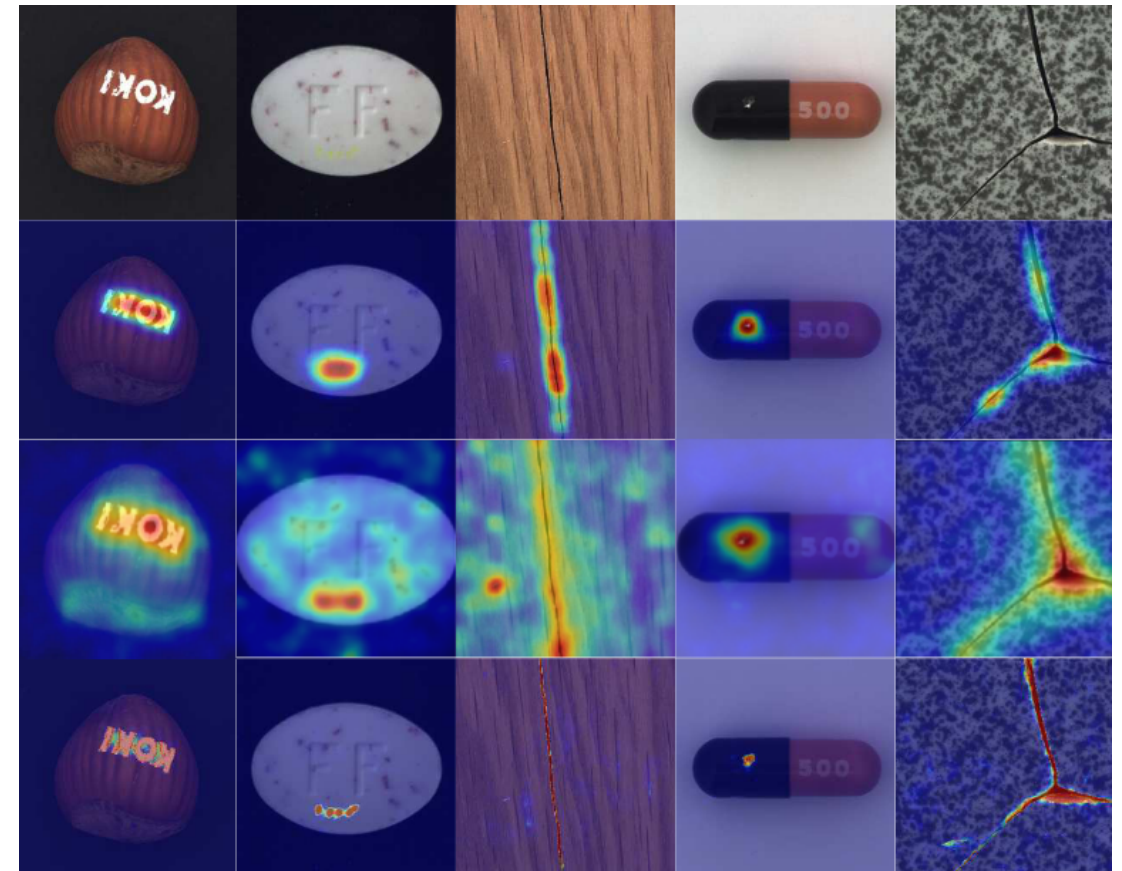
- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High-quality segmentation can help downstream tasks
 - Estimate the size of the impacted area
 - Distinguish between cracks, corrosion, or just paint
- We don't need expensive hand-labeled data



Key Findings & Application & Conclusion

The Value of DRÆM

- Outperforms unsupervised SOTA methods
- Similar performance to supervised SOTA methods
- High-quality segmentation can help downstream tasks
 - Estimate the size of the impacted area
 - Distinguish between cracks, corrosion, or just paint
- We don't need expensive hand-labeled data
- We only need just-out-of-distribution patterns for anomaly generation



The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset

The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)

The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- × Incorrect weighting of incomplete batches

The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- × Incorrect weighting of incomplete batches



Figure: Loss smoothing by weighting the last batch of each epoch.

The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- × Incorrect weighting of incomplete batches
- ✓ Complete code for training and evaluation (for main findings)

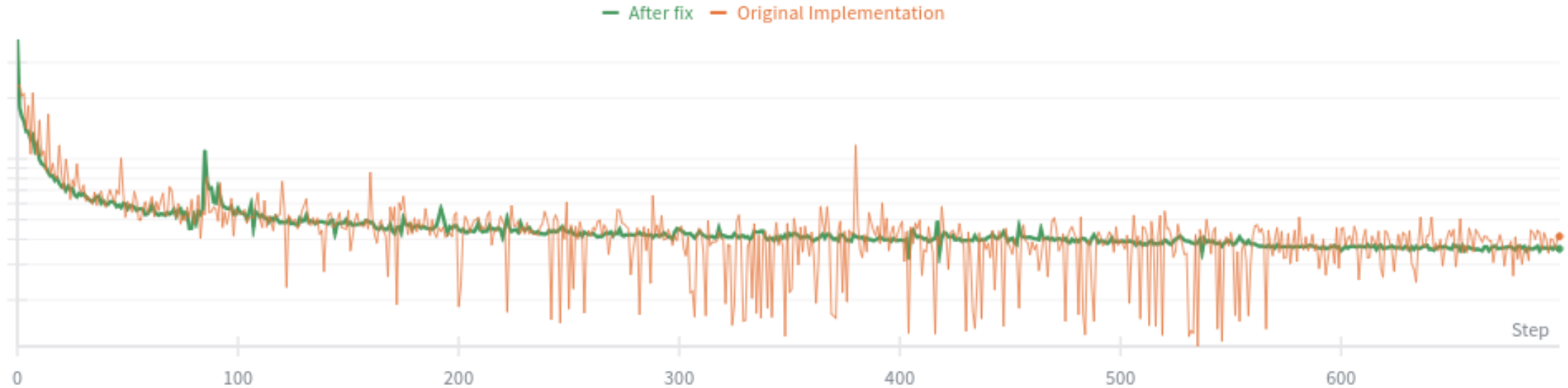


Figure: Loss smoothing by weighting the last batch of each epoch.

The Provided Code

Complete and Good?

- ✗ No code for experiments on the DAGM dataset
- ✗ Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- ✗ Incorrect weighting of incomplete batches
- ✓ Complete code for training and evaluation (for main findings)
- ✓ Well-structured code in separate files for models, loss, data loading, training and evaluation

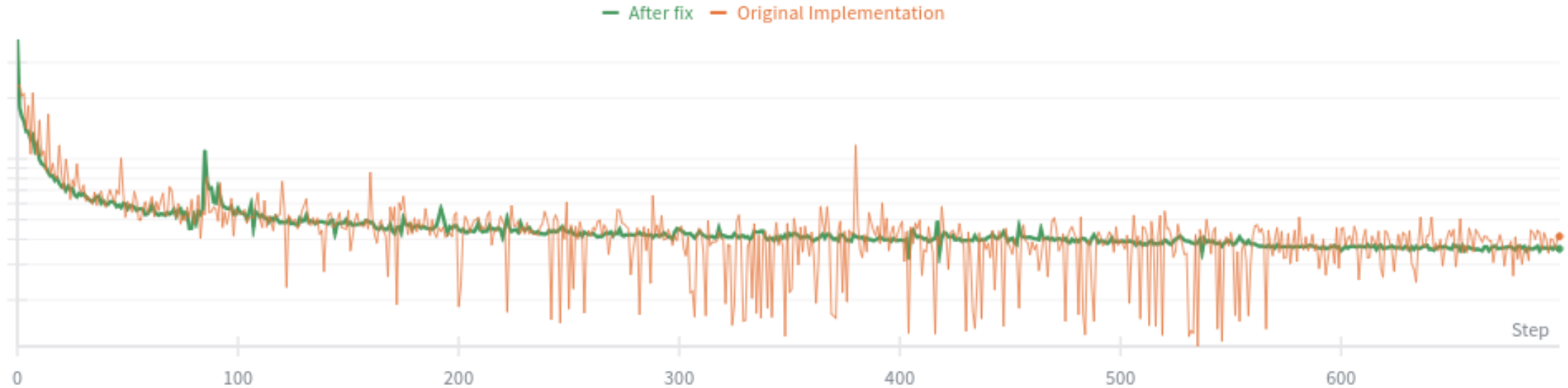


Figure: Loss smoothing by weighting the last batch of each epoch.

The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- × Incorrect weighting of incomplete batches
- ✓ Complete code for training and evaluation (for main findings)
- ✓ Well-structured code in separate files for models, loss, data loading, training and evaluation
- ✓ Well readable code

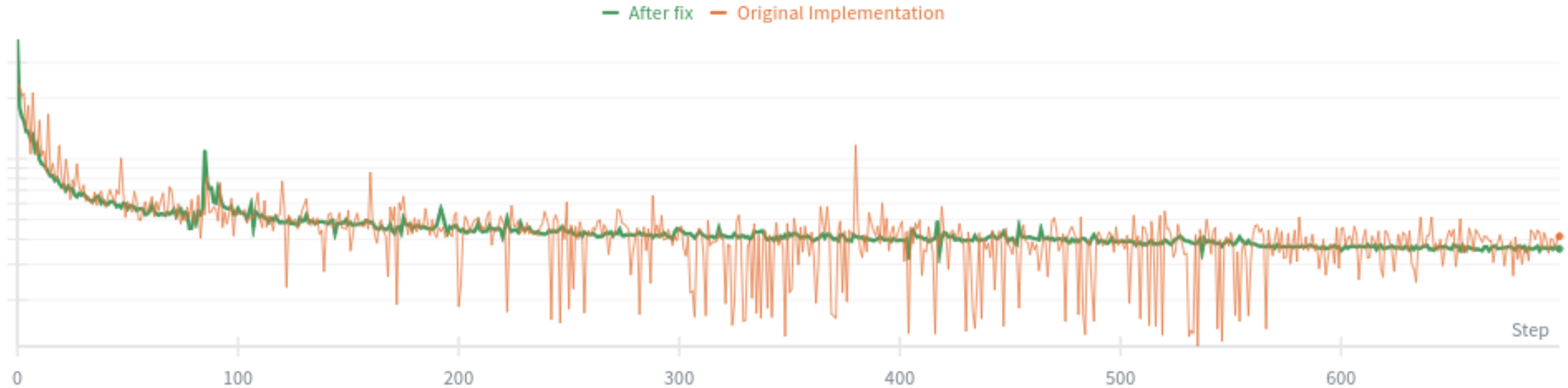


Figure: Loss smoothing by weighting the last batch of each epoch.

The Provided Code

Complete and Good?

- × No code for experiments on the DAGM dataset
- × Inconsistencies in variable meaning between code and paper (e.g., β as anomaly opacity vs. image opacity)
- × Incorrect weighting of incomplete batches
- ✓ Complete code for training and evaluation (for main findings)
- ✓ Well-structured code in separate files for models, loss, data loading, training and evaluation
- ✓ Well readable code
- ✓ Download script for MVTec and DTD datasets

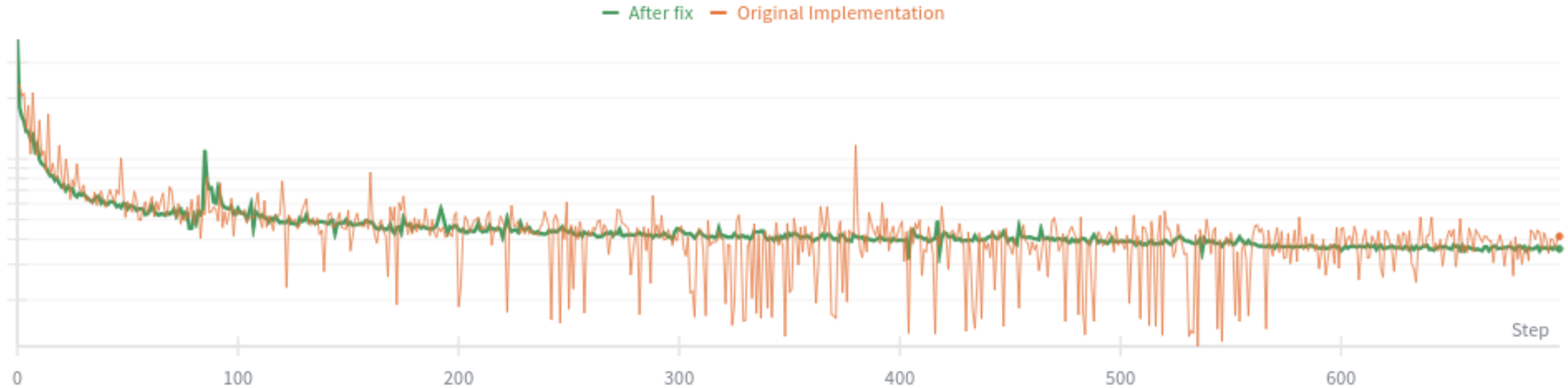


Figure: Loss smoothing by weighting the last batch of each epoch.

-
- Ablation studies in the paper show that adding transparency to anomalies has a huge impact

-
- Ablation studies in the paper show that adding transparency to anomalies has a huge impact
 - The original implementation samples a single $\beta \in [0.1, 1]$ for the whole image (uniform β)

- Ablation studies in the paper show that adding transparency to anomalies has a huge impact
- The original implementation samples a single $\beta \in [0.1, 1]$ for the whole image (uniform β)
- I added four different blending methods:

- Ablation studies in the paper show that adding transparency to anomalies has a huge impact
- The original implementation samples a single $\beta \in [0.1, 1]$ for the whole image (uniform β)
- I added four different blending methods:

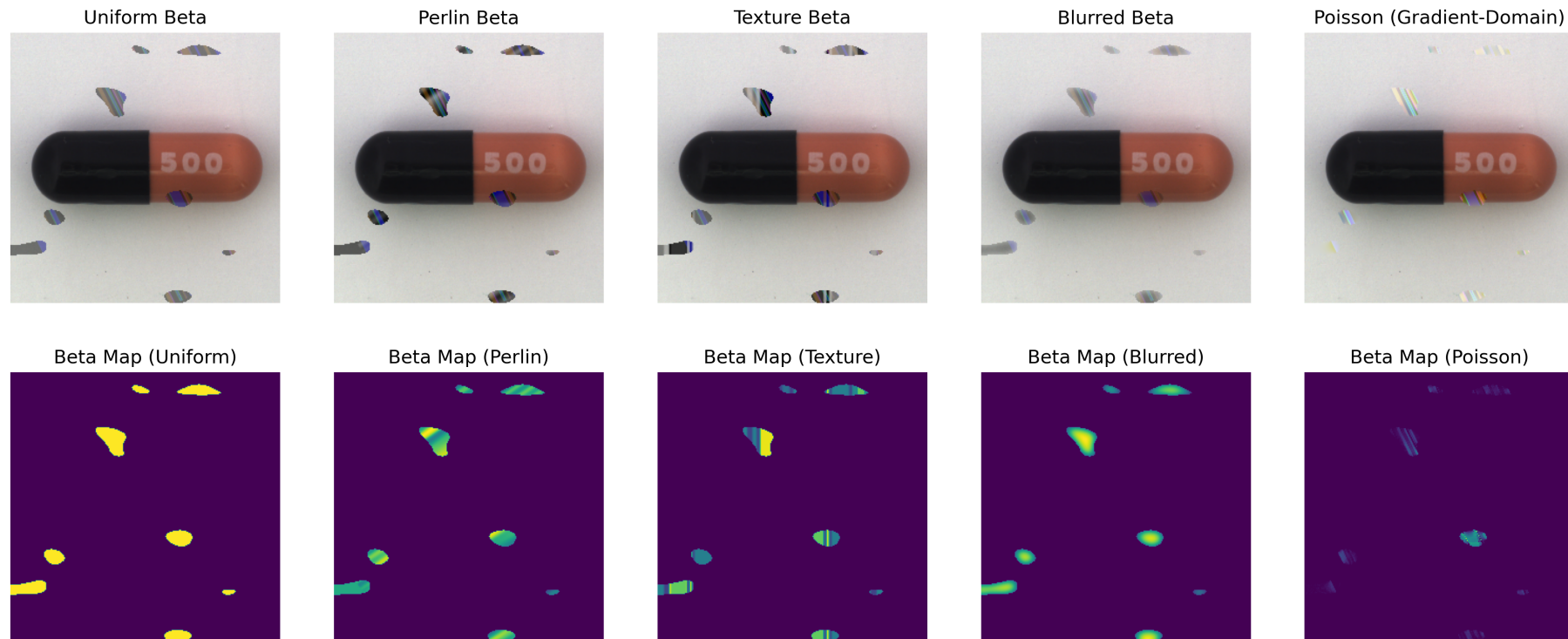


Figure: Different blending methods (from left to right): uniform β , Perlin- β , texture- β , blurred β , Poisson Image Interpolation.

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	68.4%
DRÆM-Blurred	96.4%	96.0%	65.8%
DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>

Table: Results on the MVTec dataset for different blending methods.

My Experiments

Performance Comparison

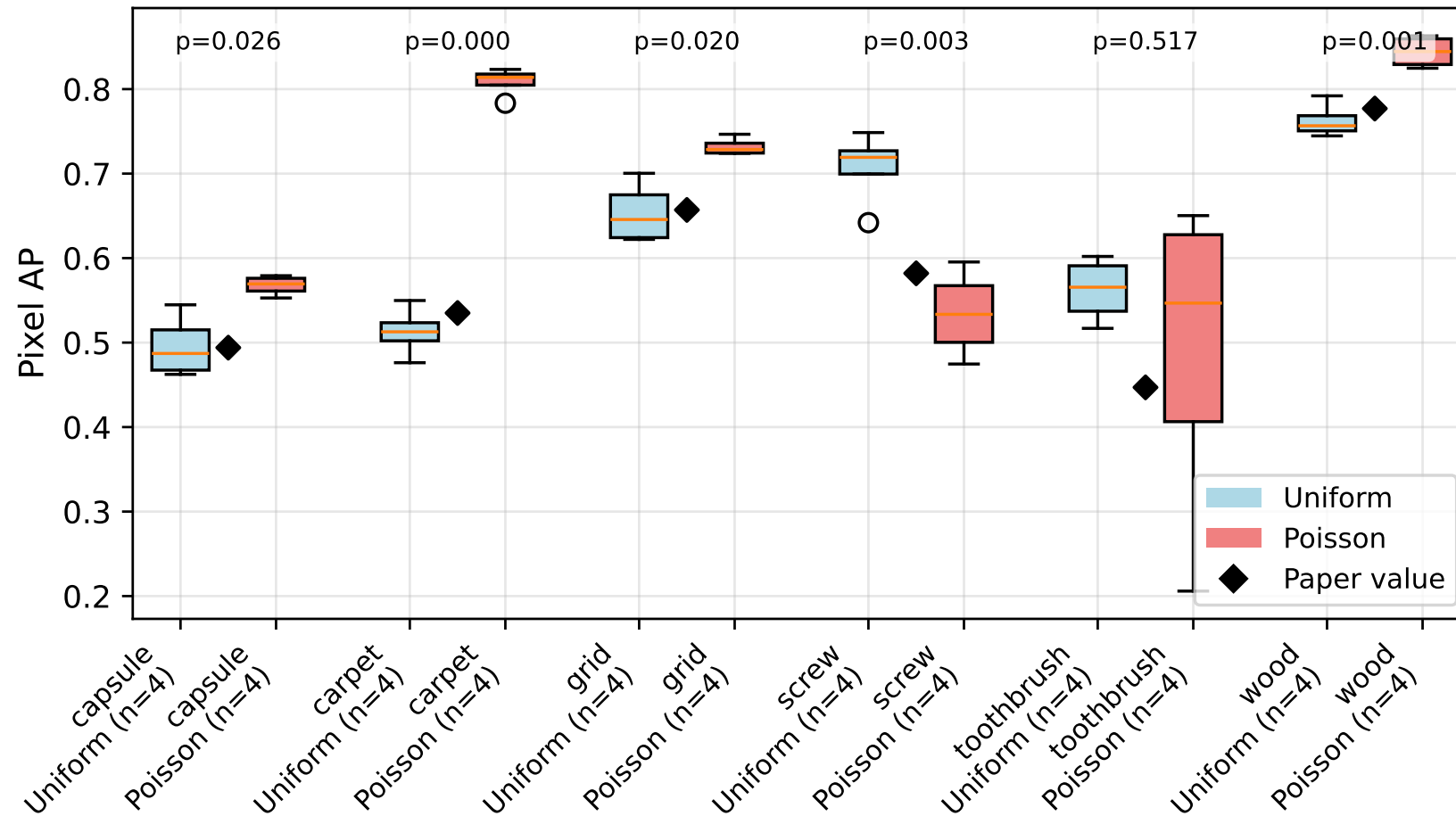


Figure: Pixel AP: Uniform blending vs Poisson image interpolation for selected object types

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	68.4%
DRÆM-Blurred	96.4%	96.0%	65.8%
DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>

Table: Results on the MVTec dataset for different blending methods.

Method	Image AUROC $\uparrow$	Pixel AUROC $\uparrow$	Pixel AP $\uparrow$
DRÆM-Uniform	98.0%	97.3%	68.4%
DRÆM-Blurred	96.4%	96.0%	65.8%
DRÆM-Texture	98.0%	97.2%	69.1%
DRÆM-Perlin	98.2%	97.3%	69.5%
DRÆM-Poisson	<u>99.1%</u>	<u>97.8%</u>	<u>71.6%</u>
DRÆM-Poisson+Perlin	99.2%	98.0%	73.6%

Table: Results on the MVTec dataset for different blending methods.

Thank you for your attention.

The background of the slide is a solid blue color. In the lower half, there is a series of concentric, curved lines that create a sense of depth and movement, resembling ripples on water or a stylized landscape.